

PROJECT 3 — PRECISION BEARING HOUSING

Project Type: CAD / 3D Modelling Capstone

Client Brief

Design a two-piece precision bearing housing consisting of a lower body and removable upper cap. Produce a manufacturing drawing package using functional datums, dimensional tolerancing and appropriate GD&T.;

Design Specification

- Bearing bore: nominal Ø40 mm
- Shaft passage: nominal Ø25 mm
- Mounting holes: nominal Ø8 mm
- Housing length: approximately 100 mm
- Two-piece housing with locating/mating features
- Include appropriate sealing or interface features

Parametric Requirements

BEARING_BORE, SHAFT_DIAMETER, HOUSING_LENGTH, HOUSING_WIDTH, MOUNT_HOLE_DIA, HOLE_PATTERN, CAP_HEIGHT and interface dimensions.

Required Skills

- Precision part modelling
- Section views
- Manufacturing drawings
- Limits and fits concepts
- General and feature tolerances
- Datum feature selection
- Datum reference frame
- Flatness, perpendicularity, parallelism, position and profile where functionally appropriate
- Functional dimensioning

Client Approval Questions

- What is the primary functional datum and why?
- What surfaces establish the datum reference frame?
- Which dimensions are functionally critical?
- Why is a particular GD&T; control being applied?
- Why use position tolerance instead of only coordinate dimensions?
- What mating condition is being protected?
- Are the tolerances realistic for the assumed manufacturing process?

CAD QA / Validation

- Model validity

- Datum consistency
- GD&T-to-function; review
- Tolerance stack awareness
- Drawing/model consistency
- Section/detail view adequacy
- STEP re-import

Required Deliverables

- Native FreeCAD model (.FCStd)
- Manufacturing / engineering drawing (PDF)
- STEP export
- STL export where relevant
- Parameter/design-intent record
- CAD QA checklist/report